Manual Tracing and Performance Analysis

Input

Reference String: 1 2 3 1 4 2 5

Number of Frames: 3

FIFO

Page	F1	F2	F3	Status
1	1	-	-	Fault
2	1	2	-	Fault
3	1	2	3	Fault
1	1	2	3	Hit
4	4	2	3	Fault
2	4	2	3	Hit
5	4	5	3	Fault

Hits = 2

Faults = 5

Hit Ratio = 28.57%

Fault Ratio = 71.43%

LRU

Page	F1	F2	F3	Status
1	1	-	-	Fault
2	1	2	-	Fault
3	1	2	3	Fault
1	1	2	3	Hit
4	1	4	3	Fault
2	1	4	2	Fault
5	5	4	2	Fault

Hits = 1

Faults = 6

Hit Ratio = 14.29%

Fault Ratio = 85.71%

Optimal

Page	F1	F2	F3	Status
1	1	-	-	Fault
2	1	2	-	Fault
3	1	2	3	Fault
1	1	2	3	Hit
4	4	2	3	Fault
2	4	2	3	Hit
5	5	2	3	Fault

Hits = 2

Faults = 5

Hit Ratio = 28.57%

Fault Ratio = 71.43%

Comparison

Algorithm	Hits	Faults
FIFO	2	5
LRU	1	6
Optimal	2	5

Analysis

FIFO and Optimal produce five page faults for this sample. LRU produces six page faults. Optimal is a theoretical benchmark that uses future page references to minimize faults. Results may differ for other inputs.

Conclusion

This manual tracing demonstrates how FIFO, LRU, and Optimal choose pages for replacement. It also shows how page hits, page faults, and their ratios are calculated.